

Brian Wong

Senior Software Engineer

Bellaire, TX, 77401 | 281-699-4292 | bw0768716@gmail.com

Summary

Senior Software Engineer with over 10 years of expertise in full-stack development, specializing in scalable and efficient backend systems. Proven leader in driving complex projects, mentoring junior developers using slack or zoom, and fostering collaboration to ensure successful outcomes. Adept at delivering high-performance solutions and guiding teams through critical technical challenges.

Work Experience

Abridge

Pittsburgh, PA

Senior Software Engineer

10/2018 - Present

- Developed APIs and data pipelines in **Python**, **Go**, and **FastAPI**, powering secure clinical note generation and provider workflow tools.
- Designed and maintained **real-time voice ingestion pipelines** using **WebRTC**, **Twilio Voice**, and **Kafka**, enabling ambient listening across mobile and EHR-integrated platforms.
- Spearheaded integration of **custom NLP and speech models**, including transformer-based medical summarizers (BERT, T5) and **custom speech-to-text** tuning, improving medical concept extraction accuracy.
- Advocated for and led the adoption of **voice SDKs** across mobile and web, enabling seamless integration into third-party EHR systems and telehealth tools.
- Implemented **low-latency audio streaming** architectures using **gRPC**, **Redis Pub/Sub**, and **WebSockets**, ensuring smooth transcription and summarization during live consultations.
- Refactored legacy voice processing modules into modular microservices, improving maintainability and supporting multi-language deployments.
- Collaborated with ML researchers to deploy medical language models via **TorchServe** and **ONNX**, with model lifecycle management through **MLflow**.
- Participated in system-level design discussions for multi-region replication, autoscaling, and queue handling using **PostgreSQL**, **Kubernetes**, and **AWS SQS**.
- Designed robust data anonymization layers using **NER-based redaction** and **de-identification heuristics**, ensuring HIPAA and GDPR compliance.
- Mentored junior engineers and led internal architecture reviews on topics such as **voice UX**, **edge inference**, and **AI observability**.

Walmart

Sunnyvale, CA

Full Stack Developer

10/2015 - 8/2018

- Contributed to internal frameworks in **Go**, **Python** and **TypeScript** that standardized backend service scaffolding, reducing onboarding time and development friction.

- Directed core platform initiatives across a high-performing software development team supporting Walmart's e-commerce backend, serving 100M+ users globally.
- Built a dynamic and responsive frontend with **React**, **Redux**, and **Material-UI**, improving user experience and accessibility for healthcare professionals.
- Developed **low-latency request classification** and routing logic using **Google AI Platform** and **Go**, reducing API tail latency from ~850ms to under 500ms in high-load environments.
- Implemented **caching** with **Redis** to speed up repeated API calls and used **Rate Limiting** on critical endpoints to prevent DDoS attacks and control API abuse.
- Enhanced throughput and reduced cross-region latency through intelligent request routing, advanced caching strategies, and **load balancing** with **Envoy** and internal mesh services.
- Orchestrated critical refactoring of legacy services into containerized **Kubernetes** deployments, introducing tighter service boundaries and platform consistency.
- Tuned **JVM-based services** for high concurrency and low GC pause times, applying deep profiling and performance tuning in **Java**, which reduced p99 response times during flash sales by over 30%.
- Built **real-time** data pipelines from production **PostgreSQL** to **Supabase** and **BigQuery** for analytics using **CDC** (Change Data Capture), ensuring dashboards remained accurate within seconds.
- Advocated for and led the integration of **voice SDKs** across mobile and web platforms using **WebRTC**, **Twilio**, and custom **STT/TTS modules**, enabling seamless conversational interfaces.
- Structured the codebase into modular, reusable components to enhance **maintainability** and **scalability**.
- Mentored junior developers and conducted code reviews to ensure adherence to best practices in software development.

Skills & Technologies

Language: JavaScript, TypeScript, Java, PHP, Python, C, C++, C#, Go

Backend: Node.js, Express.js, Nest.js, Java, SpringBoot, .NET Core, Django, Flask, Go, Laravel, RESTful API, GraphQL, Microservices

Frontend: React.js, Next.js, React Native Angular, Vue, Redux, MobX, Context API, Css3, SCSS, Webpack, Babel, Material UI, TailwindCSS, Semantic UI, Responsive and Adaptive Design

Database: PostgreSQL, MySQL, MongoDB, SQLite, Oracle, AWS RDS / DynamoDB, Supabase

DevOps & CI/CD: Git, GitHub, Bitbucket, Jenkins, CircleCI

Cloud Services: AWS, GCP, Azure, Firebase, Docker and Kubernetes for containerization and orchestration

Others: Nginx, Redis, System Design, API Integration, OpenAI, Kafka, Apache, Healthcare, Operating Systems

Education

University of Houston

Bachelor's Degree in Computer

Houston, TX

2011 - 2015